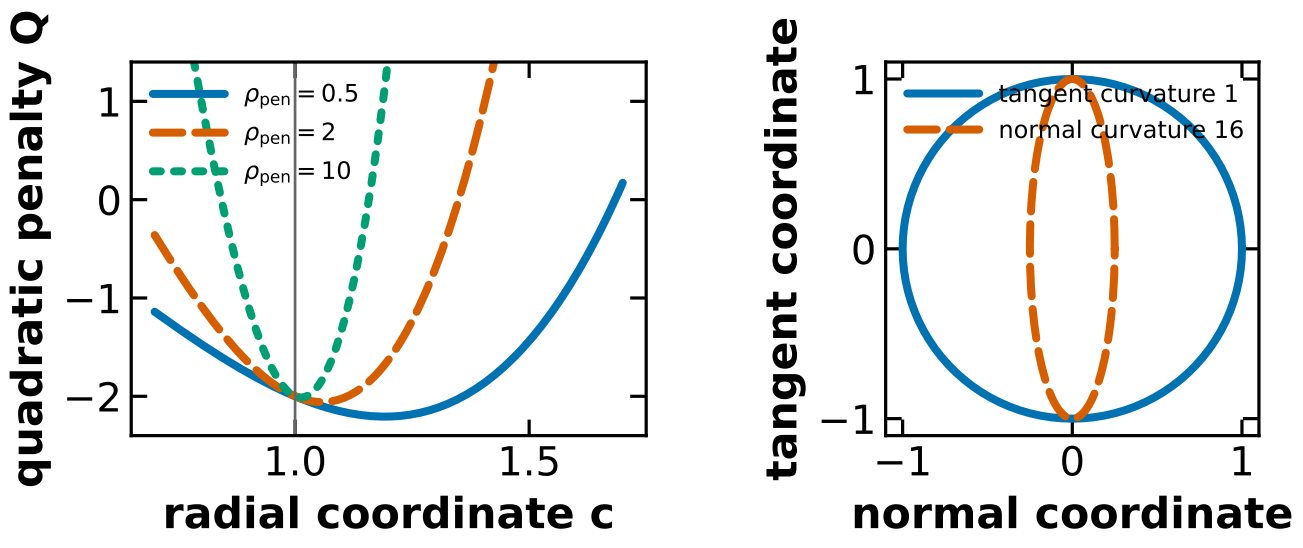
quadratic penalty Q
radial coordinate c
$\rho_{pen} = 0.5$
$\rho_{pen} = 2$
$\rho_{pen} = 10$
1.0
1.5
1
0
−1
−2
tangent coordinate
normal coordinate
tangent curvature 1
normal curvature 16
−1
0
1